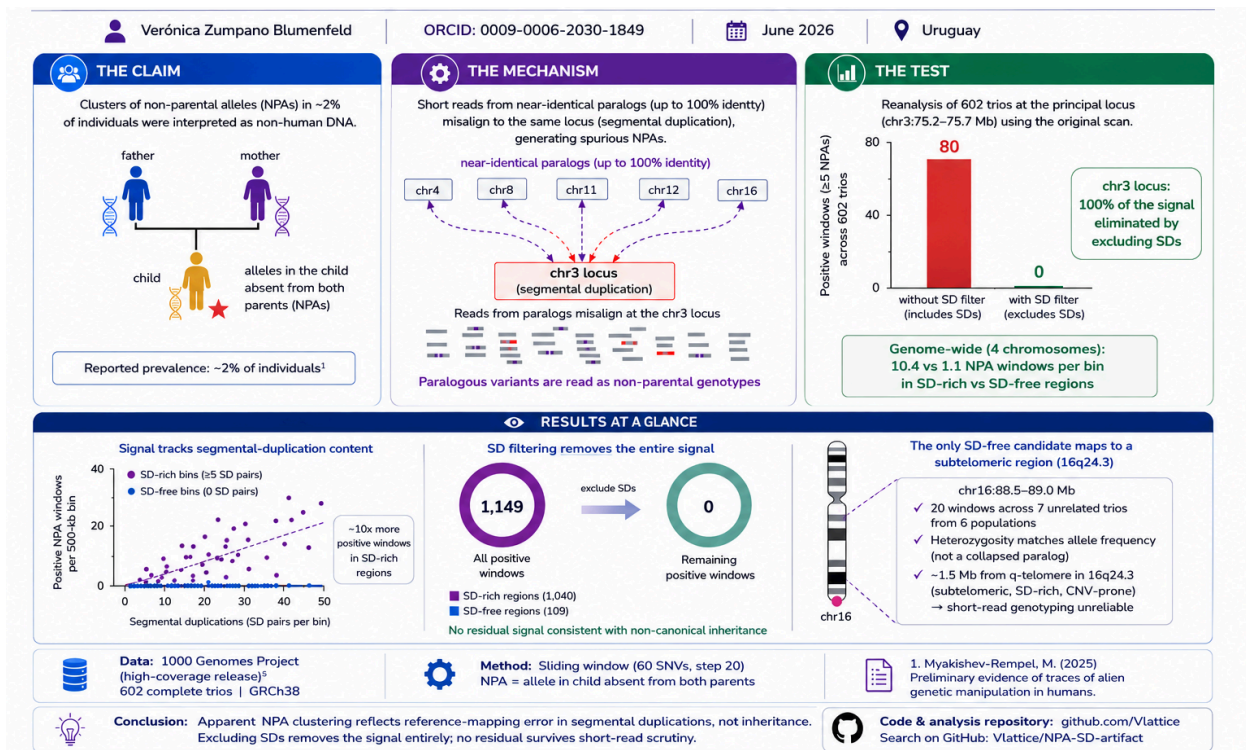


Segmental-duplication misalignment explains apparent non-parental allele clusters in short-read trio sequencing: a reanalysis.

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Graphical abstract. Reported non-parental allele (NPA) clusters coincide with segmental duplications; reads from near-identical paralogs misalign and are read as non-parental genotypes. Excluding segmental duplications removes the entire signal on chromosome 3 and the bulk of it genome-wide.

Abstract

Myakishev-Rempel recently reported that clusters of non-parental alleles (NPAs) in 1000 Genomes family trios mark traces of non-human genetic manipulation, present in roughly 2% of individuals¹. This reanalysis finds no support for this interpretation. The reported signals coincide with segmental duplications (SDs) of near-identical sequence, where short reads originating from paralogous loci misalign and generate spurious non-parental genotypes. Reprocessing 602 complete high-coverage trios at the principal reported locus (chr3:75.2–75.7 Mb) reproduced the clustering—80 windows above the detection threshold—but excluding SD intervals removed every window (100%); an SD-free control

locus (FRA3B) was negative throughout. Across 200 bins sampled from four chromosomes, NPA windows were roughly tenfold more frequent in SD-rich than in SD-free regions and scaled with SD density. The single apparent exception mapped to a subtelomeric band (16q24.3) where short-read genotyping is unreliable. The reported pattern is a consequence of reference-mapping error, not inheritance.

Main

Trio sequencing identifies a non-parental allele when a child carries an allele absent from both parents. Genuine instances are rare and arise from de novo mutation or gene conversion; the great majority of apparent cases reflect genotyping error or read misassignment². Myakishev-Rempel scanned 1000 Genomes trios for windows enriched in NPAs on chromosome 3 and interpreted recurrent clusters as engineered, non-human sequence carried by about 2% of people¹. Because that conclusion rests entirely on the premise that clustered NPAs cannot be technical, it is directly testable.

Segmental duplications—paralogous blocks longer than 1 kb that share at least 90% identity—span roughly 5% of the human genome and frequently exceed 98% identity^{3,4}. Reads from one copy align ambiguously, or incorrectly, to another, so that fixed sequence differences between paralogs appear as heterozygous or non-parental genotypes at the locus to which the reads are assigned. The three strongest clusters in the original report fall within such regions, in SD blocks of up to 100% identity with paralogs on chromosomes 4, 8, 11, 12 and 16. This predicts that the clusters should vanish once SD intervals are excluded.

The 602 complete trios of the high-coverage 1000 Genomes release⁵ were scanned with the original sliding-window NPA procedure (Methods). At chr3:75.2–75.7 Mb, the most strongly reported locus, the scan returned 80 windows above the threshold of five NPAs (Fig. 1a). Every positive window overlapped an annotated SD interval (12–45 SD pairs per window; GRCh38 GenomicSuperDup track). Excluding SD-overlapping windows eliminated all 80 positive windows (Fig. 1b). FRA3B (chr3:59.7–62.5 Mb), a comparably sized locus lacking SDs, yielded no positive windows with or without filtering.

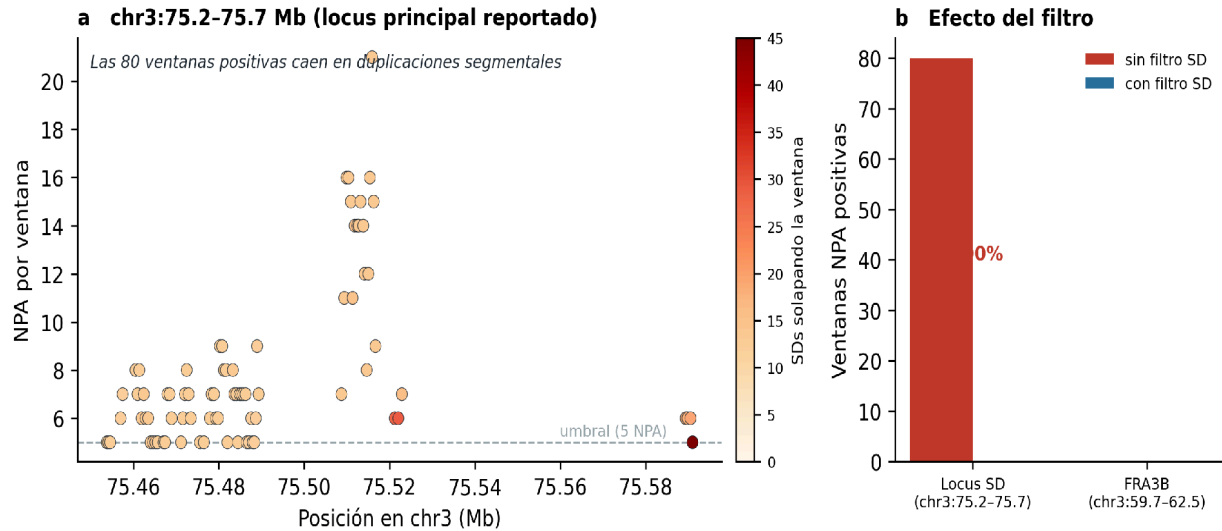


Figure 1 | The principal reported signal is an SD artifact. (a) NPA windows across chr3:75.2–75.7 Mb; every positive window overlaps a segmental duplication (colour, number of SD pairs). (b) Excluding SD-overlapping windows removes all 80 positive windows at the SD locus (–100%); the SD-free FRA3B locus is negative throughout.

To assess whether this behaviour is systematic, chromosomes 1, 2, 7 and 16 were tiled into 500-kb bins and contrasted 100 SD-rich bins (≥ 5 SD pairs) with 100 SD-free bins (zero SD pairs; Methods). SD-rich bins averaged 10.4 positive windows against 1.1 in SD-free bins (Fig. 2); 49% versus 24% contained at least one window, and the per-bin count rose with SD density. Summed across bins, SD-rich regions accounted for 1,040 of the 1,149 positive windows. The NPA signal tracks segmental-duplication content genome-wide.

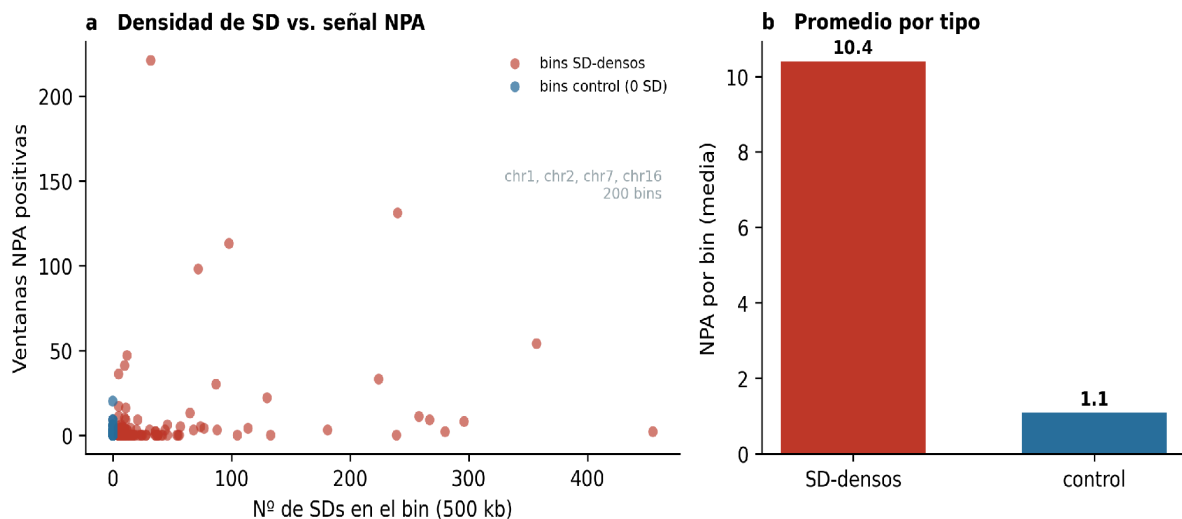


Figure 2 | NPA signal scales with segmental-duplication density. (a) Positive NPA windows per 500-kb bin versus SD count for 100 SD-rich (red) and 100 SD-free (blue) bins across four chromosomes. (b) Mean positive windows per bin by class (10.4 vs 1.1).

A single SD-free bin (chr16:88.5–89.0 Mb) carried 20 windows across seven unrelated trios from six populations—the only candidate not attributable to annotated SDs. Per-site heterozygosity matched the observed allele frequency, ruling out a collapsed paralog. The bin nonetheless lies ~1.5 Mb from the chromosome-16 q-telomere within 16q24.3, a subtelomeric band defined by recent, incompletely annotated segmental duplications and recurrent copy-number variation^{6,7}. Short-read genotyping is unreliable in subtelomeric sequence⁶, and identical positions recurring across unrelated trios are the expected outcome of a fixed reference-mapping error rather than independent transmission. This candidate is therefore considered unverifiable from short reads and is most parsimoniously explained as a further, unannotated mapping artifact.

Independent of the mapping problem, the original analysis applied no correction for multiple testing across the $\sim 5.8 \times 10^7$ window–trio comparisons implied by its design (a Bonferroni threshold near 8.6×10^{-10}), confined the empirical scan to chromosome 3 (~2.3% of the genome) while generalizing the conclusion genome-wide, and grounded its interpretive framework in non-peer-reviewed sources. The reported clustering is fully accounted for by segmental-duplication misalignment in short-read data. Applied with an SD-aware filter, the same method leaves no residual signal consistent with non-canonical inheritance. Establishing any true non-parental allele in these regions would require long-read sequencing of the relevant trios.

Methods

Data. Analyses used the high-coverage 1000 Genomes release (3,202 samples, Illumina ~30×, GRCh38; statistically phased SNV/indel/SV panel)⁵. Complete mother–father–child trios (n = 602) were identified from the project pedigree.

NPA detection. For each biallelic SNV a trio was scored as carrying a non-parental allele when the child genotype included an allele present in neither parent. Variants were scanned in sliding windows of 60 consecutive SNVs advanced in steps of 20; a window was called positive for a trio when it contained at least five NPAs. These parameters reproduce the original detection procedure.

Segmental-duplication filtering. SD intervals were taken from the GRCh38 GenomicSuperDup track (54,714 intervals). A window was classified as SD-overlapping when it intersected more than two SD intervals; SD filtering excluded such windows from the positive set.

Genome-wide comparison. Chromosomes 1, 2, 7 and 16 were tiled into non-overlapping 500-kb bins. Bins with ≥ 5 SD intervals (SD-rich) and bins with zero SD intervals (SD-free) were sampled at random, 25 of each per chromosome under a fixed seed, and scanned as above.

Candidate evaluation. At chr16:88.5–89.0 Mb the per-site fraction of heterozygous genotypes across all 3,202 samples was compared with the expectation from the observed allele frequency. Genomic context was assigned from the chromosome-16 cytoband and the published characterization of subtelomeric duplication structure.

Code and data availability. Analysis scripts are available at :

<https://github.com/Vlattice/NPA-SD-artifact>. Input genotype data are public (1000 Genomes Project).

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